

Chapter 3

One-Electron atoms: Radial Wavefunctions for ns States

The section on the quantum model of hydrogen is connected with §2.1, particularly §2.12, of Foot [2].

3.1 Quantum Mechanical Model of Hydrogen

It is fairly standard to cover the hydrogen atom (or, more generally, any one-electron atom, which we will refer to as “hydrogen”) in introductory quantum mechanics, and the reader has therefore likely seen this problem treated in all its nitty-gritty with special functions and whatnot. Just to review, the Hamiltonian for the electron in the potential of the nucleus is

$$H = U + K = -Z \frac{e^2}{4\pi\epsilon_0 r} + \frac{\mathbf{p}^2}{2m} \quad (3.1)$$

where, if we want to include the finite mass of the nucleus, we would assign $m \equiv \tilde{m}$ from Eq. (2.20).

Following standard practice, we take

$$\mathbf{p} = -i\hbar\nabla \quad (3.2)$$

and therefore $p^2 = -\hbar^2\nabla^2$. We can express the Laplacian in spherical coordinates centered on the nucleus,

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial \theta} \left(\sin(\theta) \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2}{\partial \phi^2}. \quad (3.3)$$

At this point, the usual path forward is to dive into partial differential equations and special functions and find that solutions have things like spherical harmonics and generalized Laguerre polynomials, and I am going to trust that you have seen that version of this story before. The solution is factored into the product of a radial solution, $R(r)$, and an angular solution, $A(\theta, \phi)$ and substituted into the time-independent Schrödinger equation, $H\psi(r, \theta, \phi) = E\psi(r, \theta, \phi)$.

3.2 Ansatz for Radial Solutions

What we're going to do instead is to try to avoid all the fancy special functions to get a feel for *why* the solutions have the features that they do. In particular (and this follows rationale I first saw presented by Martin Karplus) we'll simplify everything by making an educated guess (or *ansatz*) for some features of the solution $\psi(r, \theta, \phi) \equiv R(r)A(\theta, \phi)$. In particular, we'll use some intuition to assume that the energy will increase with the number of nodes in the wavefunction (which is to say the number of times that $\psi(r, \theta, \phi) = 0$ as we scan each variable).

As such, the lowest-energy wavefunction will have no nodes, and we can further guess that $A(\theta, \phi) = \text{const.}$ for the ground state (recall that since the momentum operators are spatial derivatives, wiggles in the wavefunction imply momentum, so low-energy solutions will probably not be very wiggly). This allows us to drop the second and third terms in Eq. (3.3) and the time-independent Schrödinger equation reduces to the following ODE for the radial wavefunction

$$-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dR(r)}{dr} \right) - Z \frac{e^2}{4\pi\epsilon_0 r} R(r) = ER(r). \quad (3.4)$$

This is a big improvement! We no longer need to think about angular terms and we can just turn the crank on this ODE. Taking the derivatives gives us

$$\left[\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} + Z \frac{2me^2}{4\pi\epsilon_0 \hbar^2} \frac{1}{r} \right] R(r) = -\frac{2mE}{\hbar^2} R(r). \quad (3.5)$$

To help us make sense of this mess, we can now make another educated guess about the form of the solution. For instance, we certainly know that

as $r \rightarrow \infty$, we require $R(r) \rightarrow 0$. So the large- r limiting solution, $R_{\text{lim}}(r)$, should satisfy

$$\frac{d^2}{dr^2} R_{\text{lim}}(r) = -\frac{2mE}{\hbar^2} R_{\text{lim}}(r), \quad (3.6)$$

which you can see by inspection is satisfied by

$$R_{\text{lim}}(r) \propto e^{-\alpha r} \quad (3.7)$$

with

$$\alpha = \sqrt{-\frac{2mE}{\hbar^2}}. \quad (3.8)$$

(Note that this is real since E is negative. Also, α is *not* the fine structure constant – $1/\alpha$ is a length that gives the extent of the wavefunction.)

Now that we have the asymptotic solution for the radial wavefunction, how can we find the short-range behavior? Well, there's no reason to rule out the possibility that $R_{\text{lim}}(r)$ works there, too, as it's pretty well-behaved, so let's give it a try. (This is seriously how we solve differential equations, readers – guess-and-check.) Plugging (3.7) and (3.8) into (3.5) gives us

$$\begin{aligned} \left[\alpha^2 - \frac{2\alpha}{r} + Z \frac{2me^2}{4\pi\epsilon_0\hbar^2} \frac{1}{r} \right] e^{-\alpha r} &= \alpha^2 e^{-\alpha r} \\ \left[Z \frac{2me^2}{4\pi\epsilon_0\hbar^2} \frac{1}{r} - \frac{2\alpha}{r} \right] e^{-\alpha r} &= 0. \end{aligned} \quad (3.9)$$

Multiplying both sides by $re^{\alpha r}/2$ gives

$$Z \frac{me^2}{4\pi\epsilon_0\hbar^2} = \alpha. \quad (3.10)$$

Here, recalling the Bohr model, we recognize the Bohr radius,

$$a_0 \equiv \frac{4\pi\epsilon_0\hbar^2}{me^2}, \quad (3.11)$$

and we see that the exponential decay $R_{\text{lim}}(r)$ also apparently captures the short-range physics (which we expect to be characterized by a length scale a_0/Z):

$$R(r) \propto e^{-Zr/a_0}. \quad (3.12)$$

Okay, if we really believe that we've skipped to the answer, we should ask what energy our solution has. Combining (3.8) and (3.10) gives

$$\begin{aligned} -\frac{2mE}{\hbar^2} &= \frac{Z^2}{a_0} \frac{me^2}{4\pi\epsilon_0\hbar^2} \\ E &= -Z^2 \frac{e^2}{8\pi\epsilon_0 a_0}, \end{aligned} \quad (3.13)$$

which is the energy of the ground state of the Bohr model, essentially confirming the validity of this solution since it matches the experimental value.

Just for completeness, we can normalize the whole wavefunction to get our full ground-state solution,

$$\psi_{100}(r, \theta, \phi) = \sqrt{\frac{Z^3}{\pi a_0^3}} e^{-Zr/a_0}. \quad (3.14)$$

3.2.1 Derived Quantities

Now that we have the full solution to the (non-relativistic) quantum model of the hydrogen atom in its ground state, we can use the wavefunction (3.14) to answer some quantitative questions.

For example, what is the average separation between the electron and the nucleus? Here we can compute the expectation value directly,

$$\begin{aligned} \langle r \rangle &= \langle \psi_{100} | r | \psi_{100} \rangle \\ &= \int d^3\mathbf{r} \, \psi_{100}^*(\mathbf{r}) \, r \, \psi_{100}(\mathbf{r}) \\ &= \frac{Z^3}{\pi a_0^3} 4\pi \int_0^\infty dr \, r^3 e^{-2Zr/a_0} \\ &= \frac{3}{2} \frac{a_0}{Z}. \end{aligned} \quad (3.15)$$

A slightly different question (that at first glance may seem equivalent, though it is not) is what is the *most likely* separation between the electron and the nucleus, r_m ? For this, we seek the maximum value of the probability density, $\mathcal{P} = 4\pi r^2 |\psi|^2$, found by solving

$$\left. \frac{d\mathcal{P}}{dr} \right|_{r_m} = 0 = 8\pi r_m e^{-2Zr_m/a_0} - Z \frac{8\pi}{a_0} r_m^2 e^{-2Zr_m/a_0}. \quad (3.16)$$

This yields

$$r_m = \frac{a_0}{Z}. \quad (3.17)$$

3.3 Radial Wavefunctions for Excited s states

Following our approach from above, we'd expect that the first radially-excited state will have a single node in its radial wavefunction. So we will guess that the radial wavefunction is a polynomial $p_1(r)$ of order 1 in r times the exponential decay that we needed for the large- r behavior:

$$R(r) = p_1(r)e^{-\alpha r} = \left(-b_0 \frac{a_0}{Z} + b_1 r\right) e^{-\alpha r} \quad (3.18)$$

where $b_0 \frac{a_0}{Z}$ is the radial position of the node and α is again specified by Eq. (3.8). We seek the coefficients b_0 and b_1 . The radial part of the Schrödinger equation (3.5) yields

$$\left[\frac{d^2}{dr^2} + \frac{2}{r} \frac{d}{dr} + Z \frac{2}{a_0} \frac{1}{r} \right] p(r) e^{-\alpha r} = \alpha^2 p(r) e^{-\alpha r}. \quad (3.19)$$

We're going to need some derivatives, here,

$$\begin{aligned} \left[\frac{d^2}{dr^2} \right] p(r) e^{-\alpha r} &= \left[\frac{d}{dr} \left(e^{-\alpha r} \frac{d}{dr} - \alpha e^{-\alpha r} \right) \right] p(r) \\ &= \left[e^{-\alpha r} \frac{d^2}{dr^2} - \alpha e^{-\alpha r} \frac{d}{dr} - \alpha e^{-\alpha r} \frac{d}{dr} + \alpha^2 e^{-\alpha r} \right] p(r) \\ &= e^{-\alpha r} \left[\frac{d^2}{dr^2} - 2\alpha \frac{d}{dr} + \alpha^2 \right] p(r) \end{aligned} \quad (3.20)$$

which I'm leaving in a form that operates on the polynomial part so that we can reuse these results if we want to try different polynomials later. We also have

$$\left[\frac{2}{r} \frac{d}{dr} \right] p(r) e^{-\alpha r} = e^{-\alpha r} \left[\frac{2}{r} \frac{d}{dr} - \frac{2\alpha}{r} \right] p(r). \quad (3.21)$$

We can now remove the exponential factor and we have

$$\begin{aligned} \left[\frac{d^2}{dr^2} - 2\alpha \frac{d}{dr} + \alpha^2 + \frac{2}{r} \frac{d}{dr} - \frac{2\alpha}{r} + Z \frac{2}{a_0} \frac{1}{r} \right] p(r) &= \alpha^2 p(r) \\ \left[\frac{d^2}{dr^2} + \left(\frac{2}{r} - 2\alpha \right) \frac{d}{dr} + \left(2 \frac{Z}{a_0} - 2\alpha \right) \frac{1}{r} \right] p(r) &= 0. \end{aligned} \quad (3.22)$$

Plugging in $p_1(r) = (b_1 r - b_0 a_0/Z)$ gives

$$\left(\frac{2}{r} - 2\alpha\right) b_1 + \left(2\frac{Z}{a_0} - 2\alpha\right) b_1 - \left(2\frac{Z}{a_0} - 2\alpha\right) b_0 \frac{a_0}{Z} \frac{1}{r} = 0. \quad (3.23)$$

In order for this to be satisfied for all r , we require first that the terms that are independent of r sum to zero,

$$\begin{aligned} -2\alpha + 2\frac{Z}{a_0} - 2\alpha &= 0 \\ \alpha &= \frac{Z}{2a_0} \end{aligned} \quad (3.24)$$

$$\begin{aligned} \alpha^2 &= \left(\frac{Z}{2a_0}\right)^2 \\ -\frac{2mE}{\hbar^2} &= \frac{Z^2}{4a_0} \frac{me^2}{4\pi\epsilon_0\hbar^2} \\ E &= -Z^2 \frac{e^2}{8\pi\epsilon_0 a_0} \times \frac{1}{2^2}, \end{aligned} \quad (3.25)$$

which is the ground state energy times $1/2^2$. We also require that the terms proportional to $1/r$ sum to zero, which allows us to solve for b_1/b_0 :

$$\begin{aligned} \frac{2}{r} b_1 &= 2 \left(\frac{Z}{a_0} - \alpha\right) \frac{a_0}{Z} b_0 \frac{1}{r} \\ \frac{b_1}{b_0} &= \left(1 - \alpha \frac{a_0}{Z}\right) \\ &= \frac{1}{2}. \end{aligned} \quad (3.26)$$

where we have used (3.24) to get line (3.26). Putting all of this together and normalizing the wavefunction gives us

$$\psi_{200}(r, \theta, \phi) = \sqrt{\frac{Z^5}{32\pi a_0^5}} \left(r - 2\frac{a_0}{Z}\right) e^{-Zr/(2a_0)}. \quad (3.27)$$

Note how far we have managed to get without changing variables or identifying our equations as some defining differential equation for special functions! All we had to do was to use some plausible physical arguments to make educated guesses and we have arrived at the exact, non-relativistic s -state wavefunctions for the hydrogen atom.